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DATABASE ANALYSIS

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Current Trends and Future Directions of Malignancy After kidney Transplantation: A 1970-2022 Bibliometric Analysis

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Manuscript Preparation E
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Background: Malignancy after kidney transplantation (MKT) remains a leading cause of death in transplant recipients and over the past few decades there have been many reports on this topic. However, the task of extracting crucial information from intricate events poses a significant challenge in guiding clinical work. Hence, bibliometrics was employed to summarize and predict the future in this study.

Material/Methods: Reviews and articles on MKT were extracted from the Web of Science Core Collection (WoSCC) and were analyzed by the software VOSviewer, CiteSpace, Scimago Graphica, and R package Bibliometrix for bibliometric analysis.

Results: The analysis considered 5700 publications from 28 647 authors and 4924 institutions across 100 countries, spanning the years 1970-2022. Reference co-citation analysis showed that "renal cell carcinoma", "skin cancer", "post-transplant lymphoproliferative disorder" and "COVID-19 vaccine" were research hotspots. Keywords that co-occurred early were "immunosuppressant", "cancer", "Epstein-Barr virus", "squamous cell carcinoma", and "infection", etc., while "impact","risk factor", "outcomes", "mortality", "management" frequently co-occurred later. From 2020 to 2022, newly emerging keywords such as "SARS-CoV-2" and "COVID-19", together with citation bursts for "immune checkpoint inhibitors" and "ipilimumab," were observed.

Conclusions: The focus of MKT-related studies has evolved from exploring the spectrum, risk factors, and outcomes of MKT, to examining the pathogenesis, individualized screening, prevention, and treatment, including appropriate use of immune checkpoint inhibitors. Reports of renal transplant recipients infected with SARS-CoV-2 or COVID-19 have also gained attention since 2019. These suggest that individualized management remains a frontier for research and a future direction in MKT topics.

Keywords: **Kidney Transplantation • Bibliometrics • Immune Checkpoint Inhibitors • Neoplasms**

Abbreviations: **COVID-19** – coronavirus disease 2019; **ICI** – immune checkpoint inhibitor; **KTRs** – kidney transplant recipients; **LLR** – logic-likelihood ratio; **MKT** – malignancy after kidney transplantation; **PTLD** – post-transplant lymphoproliferative disorder; **SARS-COV-2** – severe acute respiratory syndrome coronavirus 2

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Background

Kidney transplantation is considered the optimal treatment option for patients with end-stage renal disease as it enhances the quality of life and survival expectancy due to modern advancements in surgical techniques and immunosuppressive regimens. Nonetheless, long-term use of immunosuppressive agents and infection with oncogenic viruses increase the risk of de novo malignancies in renal transplant recipients compared to the general population [1]. Several observational studies have demonstrated that after adjusting for age and sex, cancer incidence in kidney transplant recipients (KTRs) is 2-4 times higher than in the general population [2]. The cumulative incidence of malignancies after renal transplantation (MKT) has been estimated to be 4-5% after 5 years, 10% after 10 years, and more than 25% after 20 years [3]. Consequently, MKT has become the second most frequent cause of death following cardiovascular disease [4], accounting for 8-10% of all deaths in KTRs [5]. Moreover, a nationwide nested case-control study of 12805 KTRs found that, among all patients who died with functional renal grafts, 56% of them passed away due to malignancies [6].

In recent years, the growing burden of MKT has prompted clinicians and researchers to investigate its prevalence, pathogenesis, risk factors, and strategies for screening, diagnosis, prophylaxis, and treatment. Although many studies on this topic have been published, most only focused on specific issues in the field of MKT. Therefore, it is of great value to thoroughly examine literature on the topic of MKT to fully understand and visualize the knowledge framework. This will provide valuable information for researchers interested in this topic. Bibliometrics is an analytical approach frequently employed to assess research trends and knowledge maps in specific disciplines [7]. This approach combines retrospective summarization and future prediction capabilities to meticulously analyze the characteristics of existing literature at a big data level. When knowledge mapping is integrated with the authenticity of scientific practice, the outcomes obtained can supplant the summarization assessments conducted through empirical accumulation. Following the process of summarization, the utilization of scientific algorithms to anticipate future research directions emerges as a more methodical approach compared to relying solely on the researcher's intuition. It is within this context that the profound importance of bibliometric analysis lies.

In this study, we employed bibliometrics to analyze substantial literature data to gain a visual and comprehensive insight into the knowledge maps, academic frontiers, and future research hotspots associated with MKT.

Material and Methods

Data Acquisition

An analysis was conducted using data from the Science Citation Index Expanded (SCI-E) of the Web of Science Core Collection (WoSCC) database, which is considered the most comprehensive and authoritative compared to other databases such as PubMed, Embase, and Scopus. The literature search was performed on February 20, 2023, with a publication retrieval range of January 1, 1970 to December 31, 2022. The search strategies were as follows: #1: TS=("cancer*" OR "neoplasm*" OR "carcinoma*" OR "malignancy*"); #2: TS=("renal transplant*" OR "kidney transplant*"); final output: #1 AND #2. Because some types of data, such as meeting abstracts, proceeding papers, editorial material, etc., cause duplicate or confused data, and the bibliometric software authorized only English language literature, we finally captured the data of English articles or review and analyzed the subject categories, contribution of countries/regions, institutions, authors, and journals. Keywords co-occurrence, reference co-citation, as well as burst strength of keywords and references, were also analyzed. The document search process is presented in **Figure 1**.

Bibliometrics and Statistical Analysis

Bibliometric analysis was performed using CiteSpace software (version 6.1.R6; the USA, downloaded from https://sourceforge.net/projects/citespace/), VOSviewer (version 1.6.18, Holland, downloaded from http://vosviewer.com), Scimago Graphica (SCImago Lab, Spain, downloaded from https://graphica.app/), and the Bibliometrix (R package 4.0.1; https://www.bibliometrix.org/home/) to map the knowledge network. Statistical analysis was completed using GraphPad Prism 9 (San Diego, CA, USA), R studio (version 4.1.2), and Microsoft Excel 365 (Microsoft Corporation; https://www.microsoft.com/en-us/microsoft-365/excel). "Hello jvenn!" online server (https://jvenn.toulouse.inrae.fr/app/example.html) was used to find the intersection part between targets.

Results

Overall Publication Performance

We analyzed 5700 MKT-related publications between 1970 and 2022 using Bibliometrix (**Figure 2A**) after document screening. These publications were issued in 1187 journals and authored by 28 647 individuals, including 221 single authors. Of the authors, 14.02% had international collaborations, and it took an average of 6.47 co-workers to complete a paper. In total, 6834 keywords were present, and 149 333 references were cited. Each article had an average lifespan of 13 years from

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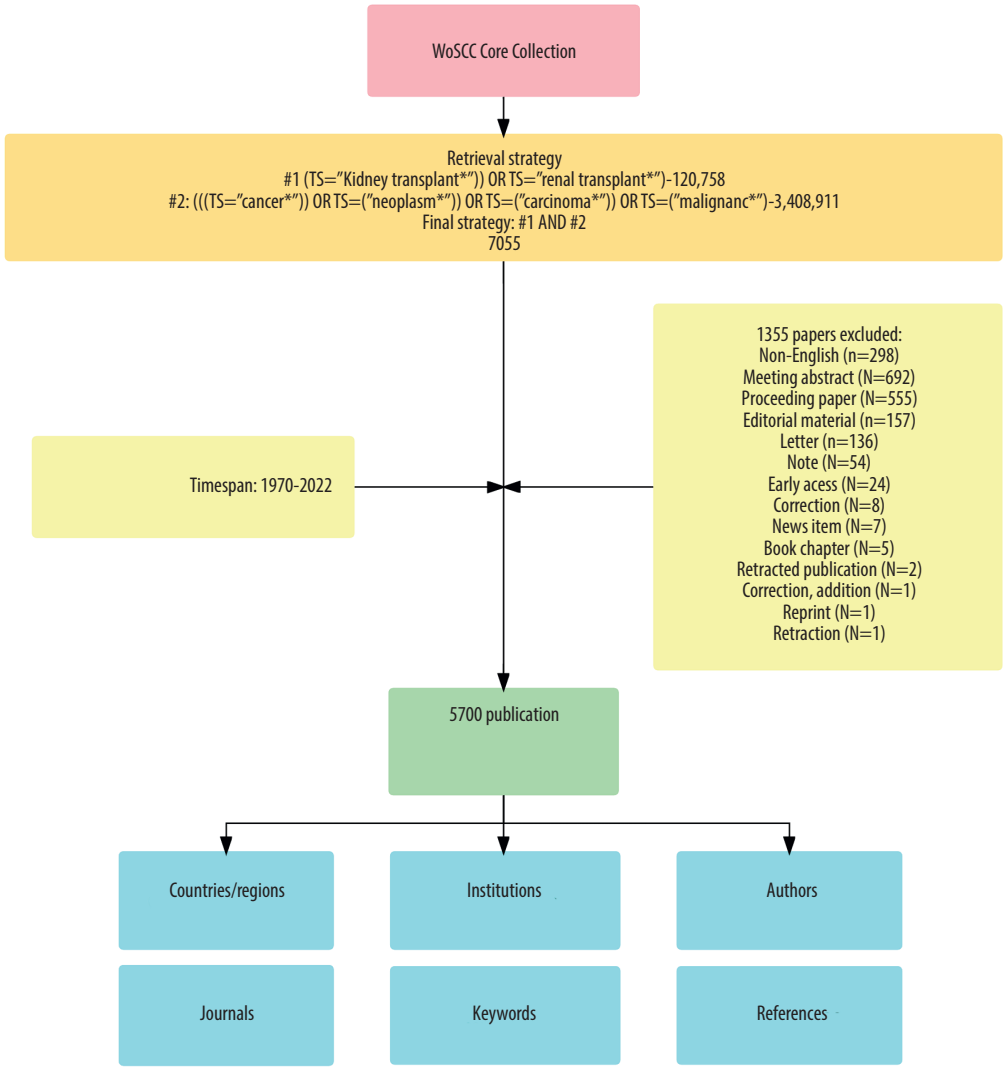


Figure 1. Flowchart illustrating the process of literature search and selection.

when it was noticed to when it became unknown, and the average number of citations was 32.28 per paper. As shown in **Figure 2B**, prior to 1992, only a few papers had been published, but the number of publications began to increase significantly from 57 in 1992 to 211 in 2022, with a peak of 291 in 2019 and an annual growth rate of 2.09%. Consequently, the number of citations rose from 1 in 1970 to 13 434 in 2021, with a slight decrease in 2022 due to a lag in statistics.

Countries/Regions of Publication and Cooperation

Using VOSviewer to construct a country/region collaboration network diagram, we imported data from all 5700 publications to analyze the global cooperation landscape. After filtering the data to include only countries with 7 or more publications, we identified 50 relevant countries out of 100. The resulting network diagram (**Figure 3A**) displayed 14 clusters for the number of international collaborations. The red sector indicated the largest cluster and included 10 countries: the USA, Australia, China, Japan, New Zealand, etc. The second-largest cluster was the green sector and included 8 countries: France, Spain, the Netherlands, India, Brazil, etc. Finally, the blue cluster had 8 countries: Germany, Poland, Switzerland, Austria, etc.

Accordingly, we presented the top 10 countries regarding the publication quantity, average citations per publication, and H-index in **Figure 3B**. The USA has the most publications (1836), followed by Germany (478) and England (426). Australia exhibits the highest average citations per publication (52.95), followed by England (50.64) and France (48.78). The USA also leads in H-index (128), followed by England (79) and Germany (77).

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Figure 2. Basic information on publications related to MKT. (**A**). Basic details of the 5700 publications included in the study. (**B**). Overall trends in annual publications and citations related to MKT from 1970 to 2022. *Analyzed by Bibliometrix and GraphPad Prism 9 software.*

Institution and Collaboration

Data from 5700 publications were imported into VOSviewer to generate a network diagram of organizational cooperation. Of the 4924 institutions, 50 met the minimum requirement of 31 publications and were included in further analysis. The results, presented in **Figure 4A**, revealed 5 clusters. For example, the red cluster included 22 collaboration groups, such as the Mayo Clinic, Harvard University, University of Southern California, etc. The green cluster comprised 17 collaboration groups, including Leiden University, Beaumont Hospital, Hannover Medical School, and others. The blue cluster comprised 8 collaboration groups: the Queen Elizabeth Hospital, The University of Sydney, Westmead Hospital, etc.

Furthermore, we presented the top 10 institutions regarding the publication quantity, average citations per publication, and H-index in **Figure 4B**. Udice-French Research Universities has the most publications (232), followed by Assistance Publique Hopitaux Paris APHP (167) and Harvard University (165). University of Sydney exhibits the highest average citations per publication (62.49), followed by Universite Paris Cite (58.57) and University of London (56.45). Udice-French Research Universities also leads in the H-index (55), followed by University of Sydney (51) and Harvard University (49).

Analysis of Co-Authorship

After analyzing 57 authors with a published lower limit of 11 papers, their co-authorship network was presented through

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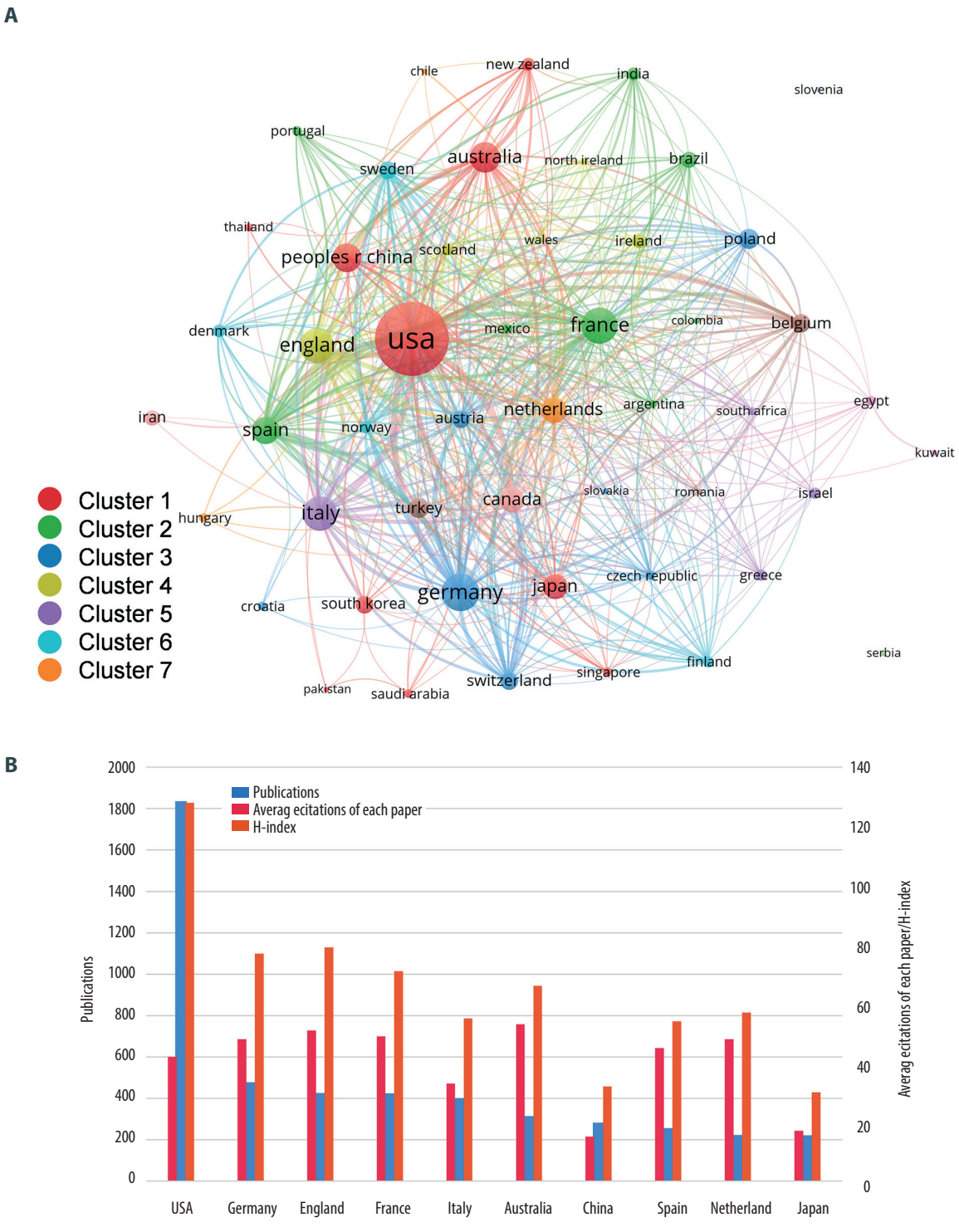


Figure 3. National science research output and international cooperation. (**A**). A visual map of the international cooperation network of countries involved in MKT. The size of the ball indicates the number of publications, while the connections indicate cooperative relationships. The countries involved in MKT-related research are categorized into 5 partner groups based on the co-authors of the 5700 publications. (**B**). The top 10 institutions in terms of publication quantity, average citations per publication, and H-index. *Analyzed by software of VOSviewer, GraphPad Prism, and Scimago Graphica.*

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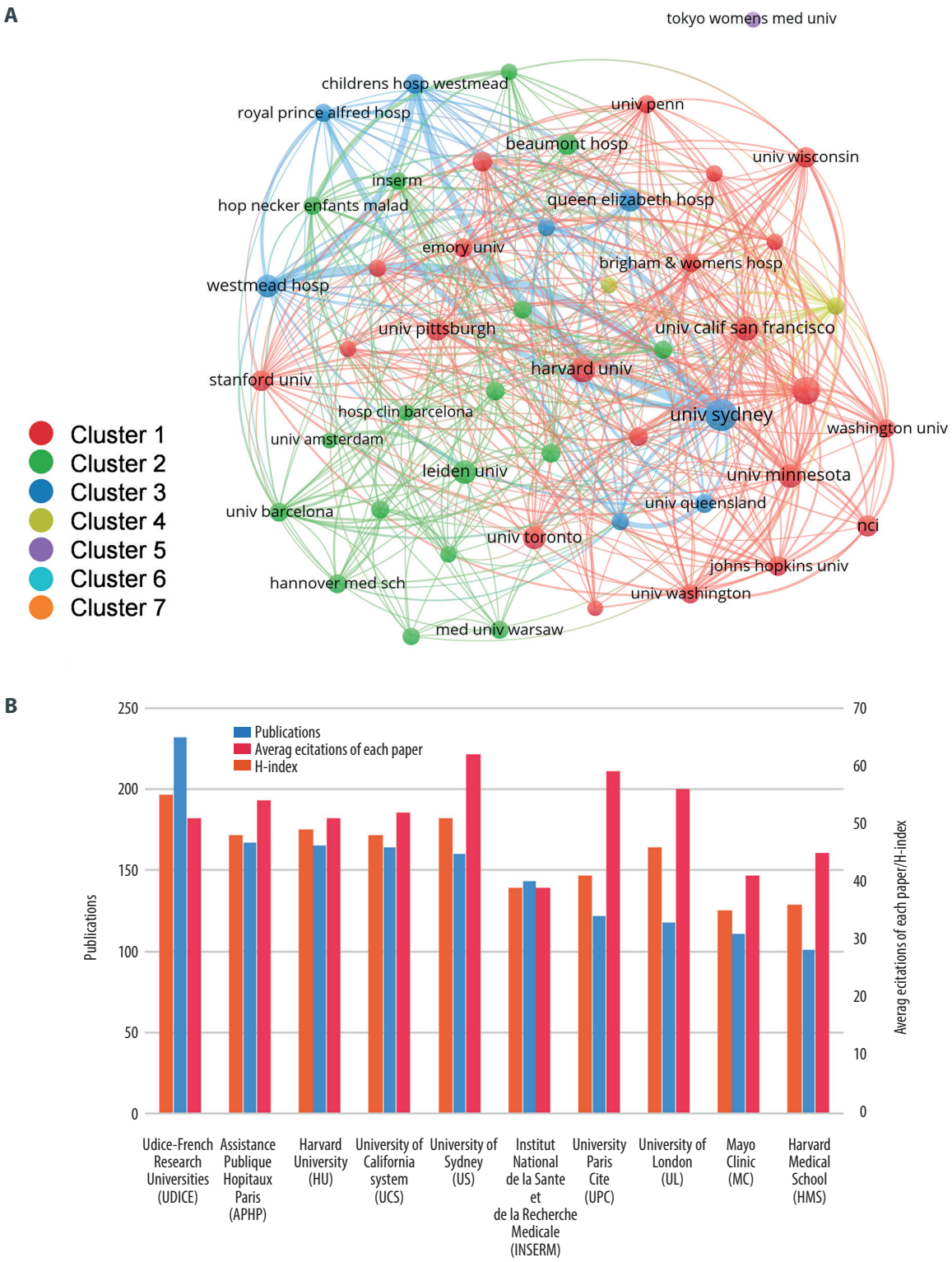


Figure 4. Institution output and collaboration. (**A**). A visual map of the current status of inter-agency cooperation in the MKT study. Different colors represent different cooperative groups, with connections indicating their collaboration. (**B**). The top 10 institutions studying MKT worldwide based on publication quantity, average citations per publication, and H-index. *Analyzed by software of VOSviewer, GraphPad Prism, and Scimago Graphica.*

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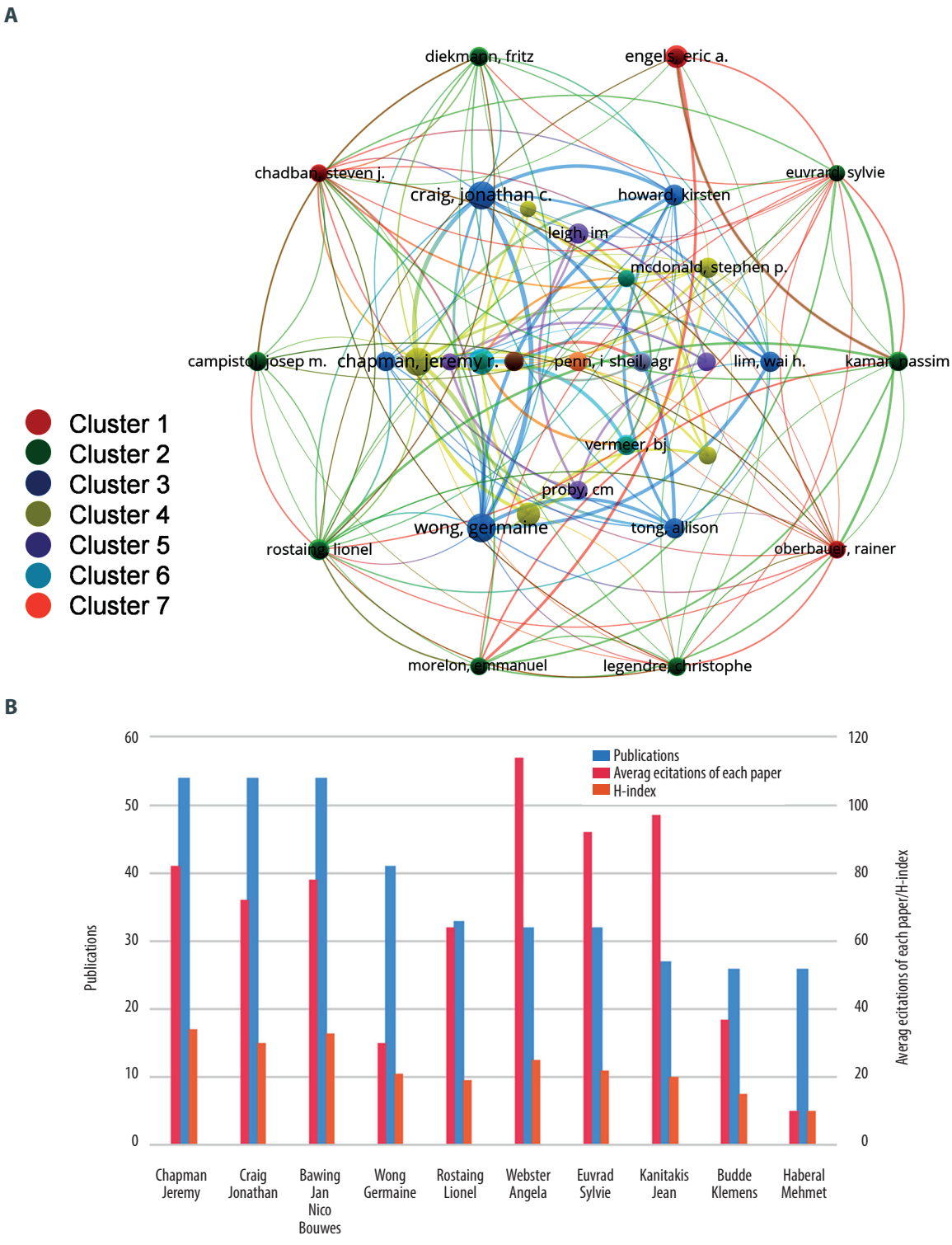


Figure 5. Co-authorship Network and top 10 authors. (**A**). A visual map of co-authorship relationships within the MKT field. Different colors represent various co-authors, and the connections denote their joint contributions. (**B**) The top 10 authors in terms of publication quantity, average citations per publication, and H-index. *Analyzed by software of Bibliometrix and GraphPad Prism 9.*

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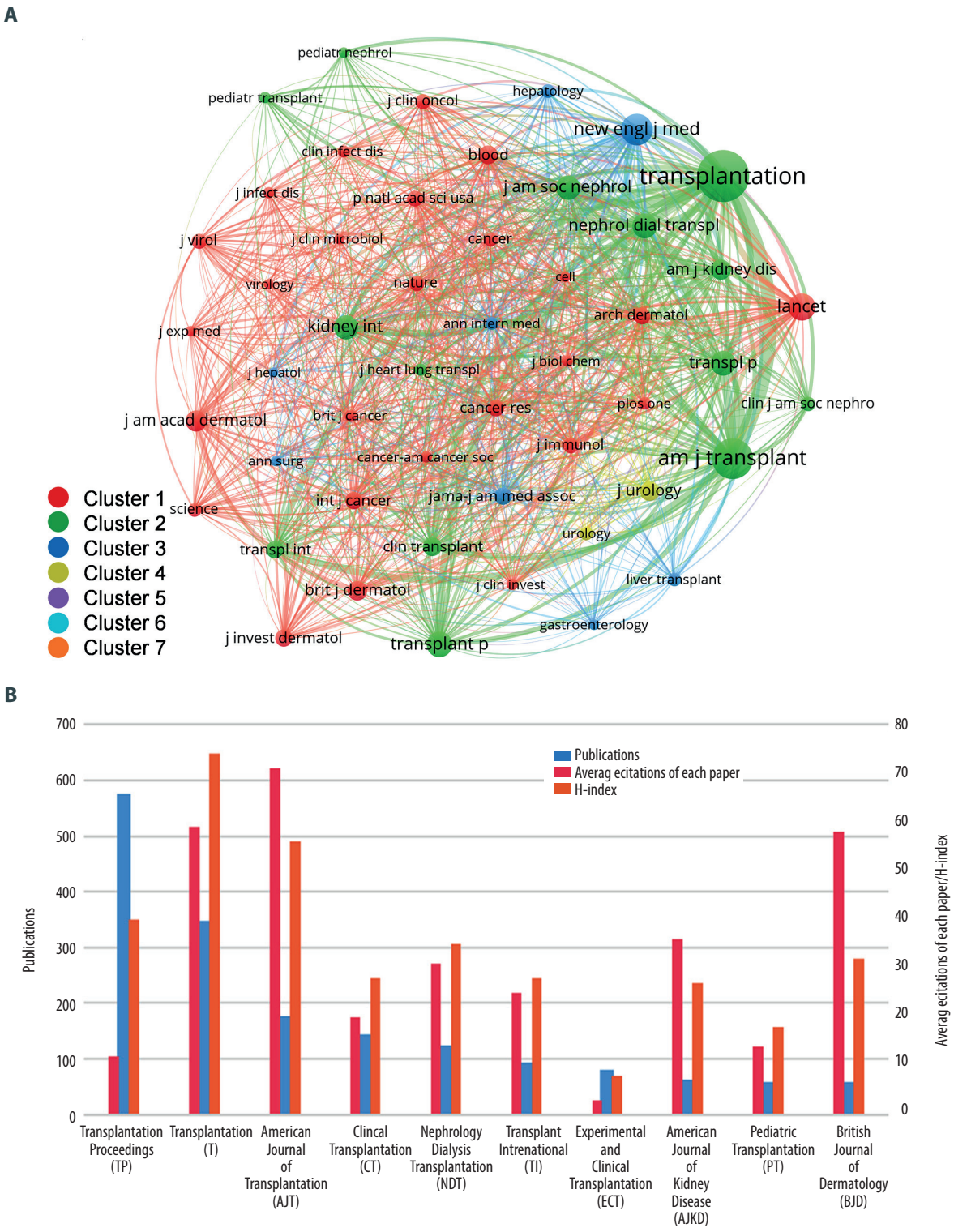


Figure 6. Journal co-citations and journal publications. (**A**). A visual map of journal co-citations has been categorized into 4 clusters. (**B**). The top 10 journals were ranked in terms of publication quantity, citation of each publication on average, and H-index. *Analyzed by software of CiteSpace and GraphPad Prism 9.*

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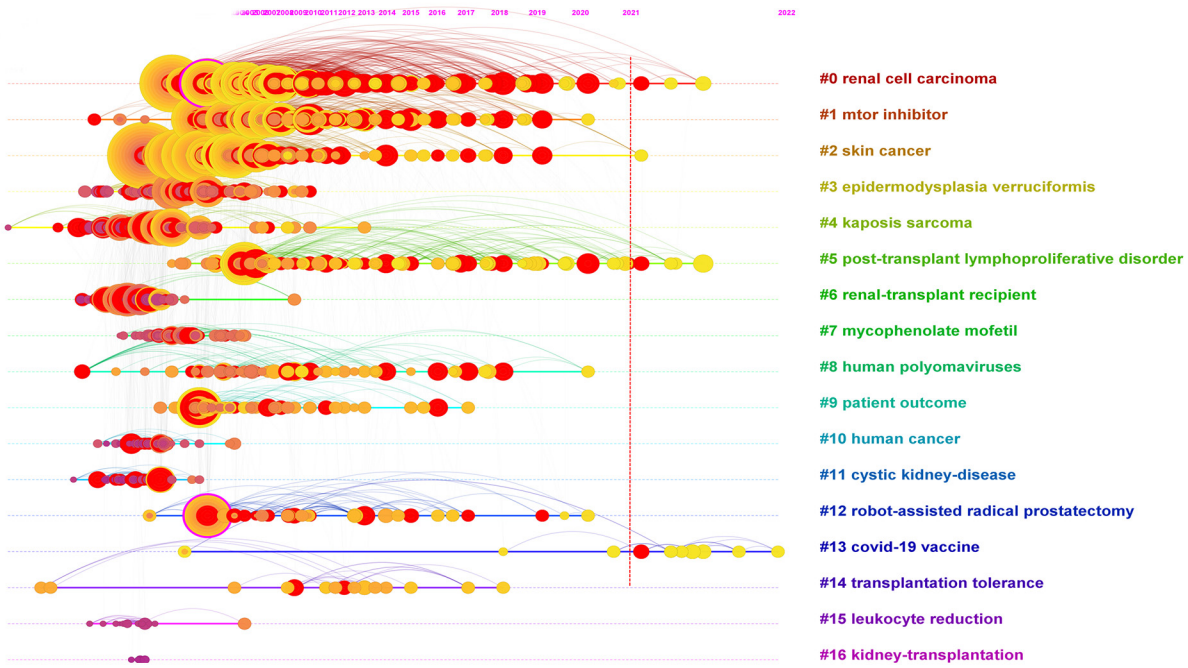


Figure 7. Analysis of reference co-citation and burst citation. Clustering analysis was conducted on the timeline view. The duration of reference co-citation was shown in 4 clusters that have crossed the time boundary of 2021, including cluster #0 ("renal cell carcinoma"), cluster #2 ("skin cancer"), cluster #5 ("post-transplant lymphoproliferative disorder"), and cluster #13 ("COVID-19 vaccine"). *Analyzed by CiteSpace software.*

18 clusters in **Figure 5A**. Additionally, the top 10 authors were ranked according to publication quantity, average citations per publication, and H-index, as shown in **Figure 5B**. Chapman Jeremy, Carig Jonathan, and Bavinck Jan Nico Bouwes have published 54 papers, respectively. Webster Angela has the highest average citations per publication (114.06), followed by Kanitakis Jean (97.19) and Euvrard Sylvie (92.28). Chapman Jeremy also leads in the H-index (34), followed by Bavinck Jan Nico Bouwes (33) and Carig Jonathan (30).

Publication Journals and Cooperation

A map depicting the co-citation network of journals was plotted, and 4 clusters were identified based on the research field and academic cross-linkage (**Figure 6A**). The red cluster contained journals that focused on research areas of dermatology, cancer, virology, and infectious diseases. The green cluster included journals interested in transplantation or nephrology. The blue cluster represented fields including liver transplantation, hepatology, gastroenterology, surgery, and medicine; the yellow cluster indicated the urology research area.

We listed the top 10 journals regarding the publication quantity, average citations per publication, and H-index in **Figure 6B**. *Transplantation Proceedings* has the most publications (575), followed by *Transplantation* (347) and *American Journal of transplantation* (176). *American Journal of transplantation* exhibits the highest average citations per publication (71.27), followed by *Transplantation* (59.32) and *British Journal of Dermatology* (58.00). *Transplantation* possesses the highest H-index (74), followed by *American Journal of transplantation* (56) and *Transplantation Proceedings* (40).

Hot and Frontier

Analysis of Reference Co-Citation and Burst Citation

We utilized the logic-likelihood ratio (LLR) to cluster the research content based on the keywords extracted from references and visualized the discipline evolution (**Figure 7**). Using a red time boundary of 2021 to identify an active co-citation status, we observed that 4 clusters emerged after 2021: cluster#0 ("renal cell carcinoma"), cluster#2 ("skin cancer"), cluster#5 ("post-transplant lymphoproliferative disorder") and cluster#13 ("COVID-19 vaccine"), which reflect research current hotspots and frontiers.

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Table 1. Top 25 References with the strongest citation bursts ended on 2022.

References title	Year	Strength	Begin	End	1991-2022
Spectrum of cancer risk among US solid organ transplant recipients	2011	66.84	2013	2022	
Comparison of the incidence of malignancy in recipients of different types of organ: A UK Registry audit	2010	25.39	2016	2022	
Primum non nocere: Is chronic kidney disease staging appropriate in living kidney transplant donors?	2009	24.49	2014	2022	
Cancer in kidney transplant recipients	2018	22.69	2019	2022	
Risk of skin cancer and other malignancies in kidney, liver, heart and lung transplant recipients 1970 to 2008 – a Swedish population-based study	2013	21.76	2014	2022	
Cancer mortality among recipients of solid-organ transplantation in Ontario, Canada	2016	19.59	2017	2022	
Cancer screening recommendations for solid organ transplant recipients: A systematic review of clinical practice guidelines	2017	18.18	2018	2022	
Malignancies after kidney transplantation: Hong Kong renal registry	2012	17.74	2016	2022	
A new equation to estimate glomerular filtration rate	2009	16.49	2016	2022	
Cancer in the transplant recipient	2013	15.58	2017	2022	
Overall and site-specific cancer mortality in patients on dialysis and after kidney transplant	2019	15.11	2019	2022	
Malignancies after renal transplantation in Taiwan: A nationwide population-based study	2012	14.9	2014	2022	
PD-1 blockade with pembrolizumab in advanced merkel-cell carcinoma	2016	14.73	2018	2022	
Immune checkpoint inhibitor related myasthenia gravis: Single center experience and systematic review of the literature	2019	13.51	2020	2022	
Pembrolizumab-related renal toxicities: Diagnosis first, treatment later	2018	12.95	2019	2022	
Long-term cancer risk of immunosuppressive regimens after kidney transplantation	2010	12.39	2013	2022	
Clinical practice guidelines on wait-listing for kidney transplantation: Consistent and equitable?	2012	12.01	2017	2022	

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Table 1 continued. Top 25 References with the strongest citation bursts ended on 2022.

References title	Year	Strength	Begin	End	1991-2022
Variation in cancer incidence among patients with ESRD during kidney function and nonfunction intervals	2016	12.01	2017	2022	
Long-term renal allograft survival in the United States: A critical reappraisal	2011	11.8	2017	2022	
Cancer incidence among US Medicare ESRD patients receiving hemodialysis, 1996-2009	2015	11.78	2018	2022	
Incidence of and risk factors for skin cancer in organ transplant recipients in the United States	2017	11.29	2018	2022	
Preserved renal-allograft function and the PD-1 pathway inhibitor Nivolumab	2017	10.8	2018	2022	
Systematic review: Kidney transplantation compared with dialysis in clinically relevant outcomes	2011	9.78	2017	2022	
Immunosuppressive drugs for kidney transplantation	2004	9.74	2014	2022	
Outcomes of solid organ transplant recipients with preexisting malignancies in remission: A systematic review and meta-analysis	2017	9.17	2019	2022	

Reference burst citation was used to identify sudden spikes in popular citations over time. **Table 1** presents the top 25 references with the strongest citation bursts ending in 2022. The blue line on the graph represents the period from 1991 to 2022, while the red line indicates the periods for each reference burst. The top-ranked reference, titled "Spectrum of Cancer Risk among US Solid Organ Transplant Recipients" [8], reports that kidney, liver, heart, or lung transplant recipients face an increased risk of cancers compared to the general population, based on an analysis of 1 757 320 solid organ transplants.

Analysis of Keywords Co-Occurrence and Burst Citation

By examining the visual patterns of keyword co-occurrence, we can gain insights into how research hotspots are distributed and evolve within a particular field. After setting a lower limit of 139 occurrences for a keyword, 50 high-frequency keywords were clustered. **Figure 8A** shows that before 2010, the frequent co-occurrence of keywords included "immunosuppressant", " therapy", "cancer", "tumor", "Epstein-Barr virus", "squamous cell carcinoma", and "infection", which were primarily included in the blue cluster. However, keywords such as "risk", "impact", "risk factor", "outcomes", "survival", "malignancies", "cancer", "mortality", and "management" became more common over time, which were mainly included in the red cluster. To further determine the research focus between 2020 and 2022, we only included 739 articles from 2020-2022 for keyword co-occurrence analysis to predict frontier hotspots. After setting a lower limit of 10 occurrences for a keyword, 115 keywords were clustered in **Figure 8B**. We targeted the following 17 keywords after taking the intersection of the 20 keywords with the highest score<Avg. citations> and the 20 keywords with the highest score<Avg. norm. citations> from the 115 keywords using a Wayne diagram: "immune checkpoint inhibitors", "ipilimumab", "antibody-mediated rejection", "allograft-rejection", "renal-transplant recipients", "acute kidney injury", "immunotherapy", "nonmelanoma skin-cancer", "solid organ transplant", "stage renal-disease", "COVID-19", "kidney-transplant", "calcineurin inhibitors", "t-cells", "solid-organ transplantation", "hemodialysis", and "experience" (**Figure 9**).

Burst detection of keywords was also used to identify research hotspots and directions over time. **Table 2** displayed the 39 keywords that remained active in citation periods up to 2022. The blue line in the graph represents the period from 1991 to

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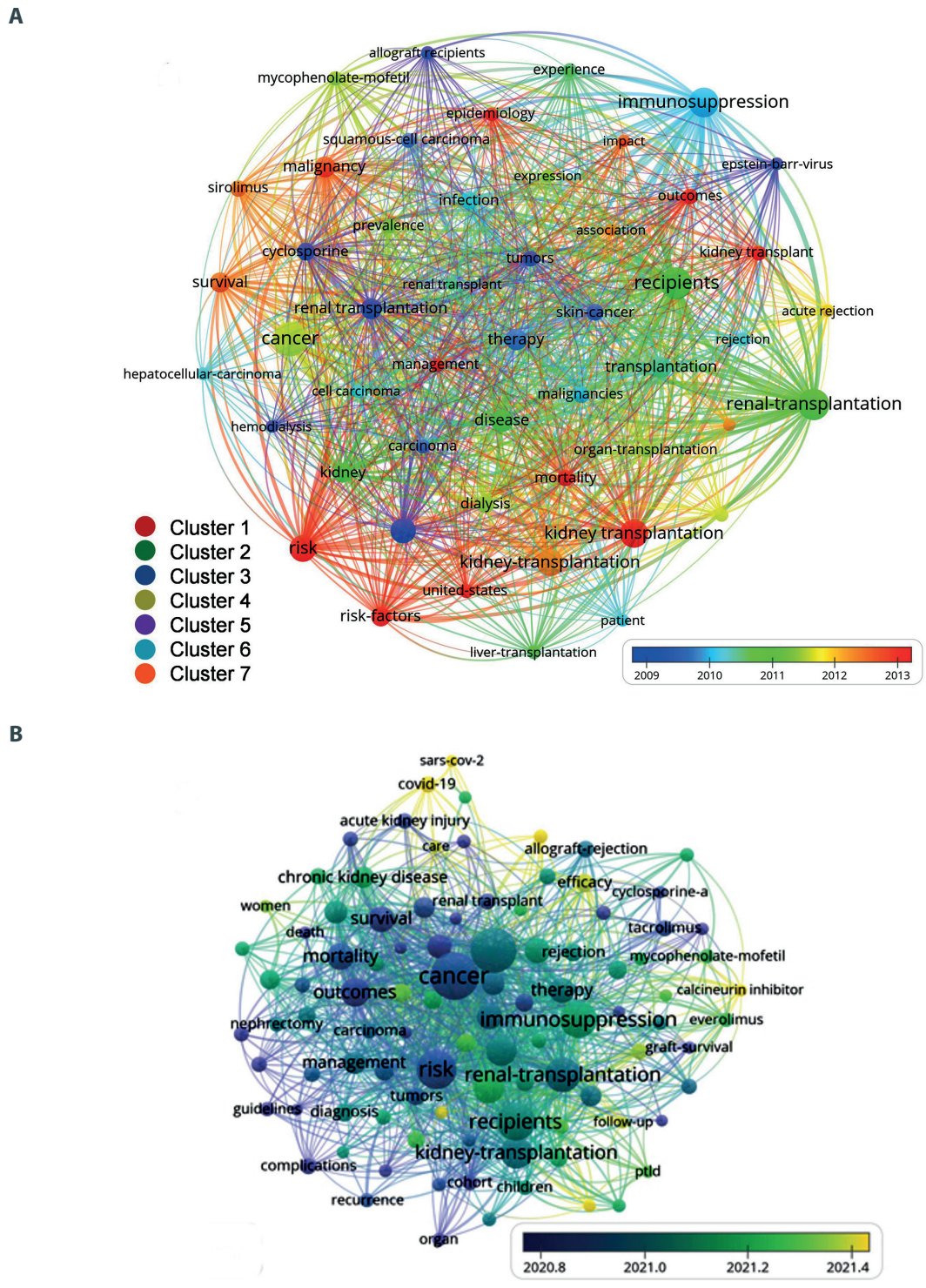


Figure 8. Analysis of keywords co-occurrence and burst citation. (**A**). Co-occurrence analysis was performed on the visual map of keyword co-occurrence, which was categorized into 7 clusters. Before 2010, keywords such as “immunosuppressant”, “therapy”, “cancer”, “tumor”, and “infection” occurred frequently. After that, keywords such as “risk”, “impact”, “risk factor”, “outcomes”, “survival”, “malignancies”, “cancer”, “mortality”, and “management” have become more common. (**B**). Co-occurrence analysis of keywords emerged between 2020 and 2022. The most newly emerging keywords are “SARS-COV-2” and “COVID-19”. *Analyzed by software CiteSpace, VOSviewer, and Microsoft Excel 365.*

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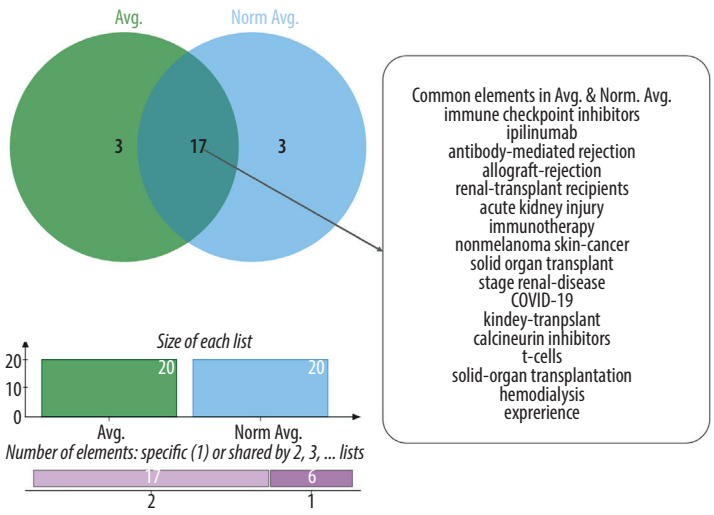


Figure 9. Wayne diagram of the intersection results of the 20 keywords with the highest score<Avg. citations> and the 20 keywords with the highest score<Avg. norm. citations> in the 2020-2022 keyword co-occurrence analysis results. 17 keywords were targeted after taking the intersection of the 20 keywords with the highest score<Avg. citations> and the 20 keywords with the highest score<Avg. norm. citations> from the 115 keywords using a Wayne diagram: "immune checkpoint inhibitors", "ipilimumab", "antibody-mediated rejection", "allograft-rejection", "renal-transplant recipients", "acute kidney injury", "immunotherapy", "nonmelanoma skin-cancer", "solid organ transplant", "stage renal-disease", "COVID-19", "kidney-transplant", "calcineurin inhibitors", "t-cells", "solid-organ transplantation", "hemodialysis", and "experience". Analyzed by software of VOSviewer, and "Hello jvenn!" online server.

2022, while the red line plots the periods of each burst keyword. Notably, with great significance, "immune checkpoint inhibitor" and "ipilimumab" emerged as new burst keywords from 2020 to 2022.

Discussion

Current Status and Trends of Research in the Field of MKT

To our knowledge, this is the first bibliometric analysis assessing the research evolution of MKT from 1970 to 2022. After excluding 1355 documents that did not meet the inclusion criteria, 5700 publications from 1187 journals in 100 countries/regions and 4924 institutions were eligible for analysis (**Figure 2A**).

The clinical application of potent immunosuppressants, such as cyclosporine and tacrolimus, in kidney transplant recipients between 1980 and 1990, has significantly prolonged the survival period of these patients [9]. This has enabled scholars to conduct more clinical research [10]. Correspondingly, there has been an increase in publications on MKT after 1990 (**Figure 2B**). This suggests that while we have made considerable strides in comprehending MKT over the past decades, many issues still require resolution. Further investigation into MKT remains a topic of widespread interest among academic researchers.

The yearly publication increase may be attributed to cooperation among countries, institutions, authors, and journal contributions. **Figure 3A** depicts the strong international collaboration between the USA and Asian countries such as China, Japan, Australia, and South Korea, as well as the international cooperation between European countries. As the country with the largest number of publications and the highest H-index, the USA is a leading country (**Figure 3B**). This could be partly because the USA has the most significant number of yearly kidney transplants worldwide and a well-established research foundation for organ transplantation [11]. Regional cooperation among institutions and collaboration among authors are also common. Among the top 10 institutions in terms of publication volume, the top 2 are from France and the remaining 5 are from the United States, with no institutions from Asia making it into the top 10 (**Figures 4, 5**). This indicates that academic institutions in Asia still need to strengthen their research on MKT.

Citation analysis of the journal is widely used in the academic field to investigate interdisciplinary structures. This helps authors to understand the influence of periodicals and areas of focus and, subsequently, choose the most appropriate journal for submission [12]. Among the top 10 journals in terms of publication quantity, 9 focus on the transplant field, indicating that these journals will likely be preferred by authors studying MKT. "*Transplantation*" and "*American Journal of Transplantation*" are the top 2 authoritative journals in the field of transplantation, with high citations and H-index. The top 10 journals ranked by publication quantity consist of 5 from the USA, 3 from England, one from Denmark, and one from Turkey, reflecting the limited influence of Asian countries in the academic field of organ transplantation. These findings suggest that the government should encourage research institutions to engage in international cooperation, thereby improving the academic level and influence and publishing high-quality papers.

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Table 2. Top 39 Keywords with the Strongest Citation Bursts ended on 2022.

Keywords	Year	Strength	Begin	End	1991-2022
Outcome	2006	23.19	**2015**	2022	
Chronic kidney disease	2006	14.7	**2018**	2022	
Mortality	1997	13.05	**2019**	2022	
Acute kidney injury	2016	10.58	**2016**	2022	
Stage renal disease	2004	9.96	**2019**	2022	
Kidney transplant	1995	8.44	**2018**	2022	
Clinical practice guideline	2018	8.2	**2018**	2022	
Allograft rejection	1991	8.11	**2018**	2022	
Guideline	2014	7.9	**2014**	2022	
Kidney transplant recipient	2007	7.38	**2016**	2022	
Kidney transplantation	2005	6.88	**2019**	2022	
Recurrence	1992	6.47	**2019**	2022	
Case report	2017	6.44	**2020**	2022	
Care	2018	6.38	**2018**	2022	
Solid organ transplant	2016	6.14	**2019**	2022	
Solid organ transplantation	2005	6.06	**2018**	2022	
Mechanism	2016	6.06	**2019**	2022	
Quality of life	2003	5.94	**2016**	2022	
Kidney disease	1994	5.52	**2018**	2022	
Management	1996	5.47	**2018**	2022	
B cell	2012	5.41	**2018**	2022	
Cancer incidence	1996	5.38	**2019**	2022	
Liver transplant	2009	5.36	**2020**	2022	
Oxidative stress	2007	5.31	**2017**	2022	
Impact	1996	5.01	**2017**	2022	
Immune checkpoint inhibitor	2020	4.85	**2020**	2022	
Systemic lupus erythematosus	2000	4.76	**2020**	2022	
Registry	2012	4.63	**2016**	2022	
Ipilimumab	2020	4.54	**2020**	2022	
Nephrectomy	2004	4.33	**2017**	2022	
Inflammation	2013	4.33	**2017**	2022	
Diagnosis	2000	4.16	**2017**	2022	
Living donor	2010	4.14	**2018**	2022	
BK polyomavirus	2017	3.95	**2017**	2022	

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Table 2 continued. Top 39 Keywords with the Strongest Citation Bursts ended on 2022.

Keywords	Year	Strength	Begin	End	1991-2022
Replication	2018	3.95	**2018**	2022	
Graft	2005	3.89	**2016**	2022	
Death	2009	3.85	**2020**	2022	
Liver	2000	3.79	**2020**	2022	
Injury	2017	3.6	**2017**	2022	

Historical Evolution of Research Content and Research Hotspots in MKT

Reference co-citation analysis can provide valuable insights into the historical evolution of research content within a particular field. As depicted in **Figure 7**, reference co-citation after 2021, which could partly indicate the research hotspots and frontiers, were clustered as "renal cell carcinoma", "skin cancer", "post-transplant lymphoproliferative disorder", and "COVID-19 vaccine". This is consistent with previous studies that have suggested that although the most common types of MKT reported in different regions and countries vary, skin cancers (especially nonmelanoma skin cancer), post-transplant lymphoproliferative disorder (PTLD), and renal cell carcinoma are the most prevalent MKTs [13-15]. As the most common immunodeficiency individuals, KTRs and patients with malignancies are at a high risk of developing severe COVID-19 and have higher mortality rates [16]. Despite the availability of safe and effective COVID-19 vaccines that protect against severe outcomes of COVID-19, especially for patients with cancer [17], it is also important to note that conventional vaccine strategies against SARS-CoV-2 may not generate sufficient immunological responses or provide adequate protection from COVID-19 for KTRs [18]. This may be one of the reasons why there is extremely limited literature on COVID-19 or COVID-19 vaccines in patients with MKT. Therefore, there is an urgent need for prospective, randomized, controlled clinical trials to thoroughly examine the immunogenicity, long-term efficacy, and safety of COVID-19 vaccines for this vulnerable population [19-22]. Consequently, new approaches are required to bolster the protective immunity conveyed by vaccination within this high-risk patient group [23].

Further evidence supporting these findings was revealed by analyzing reference burst citations, as presented in **Table 1**. Specifically, the top 25 references with the strongest citation bursts ending in 2022, which may indicate the focal points of MKT research, pertained to investigating cancer incidence and spectrum, clinical outcomes of MKT [2, 4, 24-32], risk factors [8, 33, 34], cancer screening [33] and therapy for MKT especially, immune checkpoint inhibitors treatment [35-37]. Studies have suggested that transplant-related risk factors of MKT include immunosuppressant usage (especially T cell-depleting agents), acute rejection, sensitization status, virus infection, and pre-transplant dialysis duration [38]. Furthermore, strong observational evidence is available now that chronic kidney disease (CKD), regardless of the CKD stage, is related to an increased risk of cancer and poor prognosis [39]. Considering the various factors contributing to the increased susceptibility of kidney transplant recipients (KTRs) to certain types of cancer, it is imperative to conduct large-scale prospective studies involving long-term follow-up data from patient registries to understand them better in MKT incidence, risk factors, and mortality.

Keyword co-occurrence and citation burst analyses enable us to grasp the landscape of scholarly inquiry better and identify emerging trends and themes that may require further investigation. As shown in **Figure 8A**, early studies prior to 2010 focused on fundamental knowledge related to MKT and were characterized by the keywords such as "immunosuppressant", "Epstein-Barr virus", "squamous cell carcinoma", "cancer", "tumor", and "infection". These studies demonstrated that MKT increases the risk of death in KTRs. Subsequently, the focus of research shifted towards MKT screening and patient management to improve clinical outcomes, as reflected in the keywords such as "epidemiology", "risk factor", "impact", "outcome", "survival", "mortality", "management", "SARS-COV-2", "COVID-19", among others (**Figures 8B, 9**). These keywords also appear in **Table 2** as MKT research hotspots, as evidenced by their strong citation bursts in 2022. We also observed frequent co-occurrence of keywords such as "BK polyomavirus", "B cell", "replication", "systemic lupus erythematosus", and "mechanism", indicating that the pathogenesis of MKT remains an area of active research. Despite the lack of a clear understanding of the exact mechanisms involved, it is widely accepted that long-term immunosuppression can lead to dampened immune systems, potentially resulting in cancer development through various pathways [2]. Therefore, while there is limited agreement on personalized preventive measures for MKT, minimizing risk factors, such as restricting skin exposure, dermatological screening, and prophylactic retinoid use, can help reduce the incidence of skin malignancies. Endoscopic screening for

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gastrointestinal cancer in MKRs can also lead to early identification and improved outcomes. Additionally, certain MKT has shown a positive response to antiviral treatment [40]. Thus, formulating individualized strategies for MKT prevention and screening is an urgent area of clinical research.

Emerging Research Hotspots in the Past Few Years

The most noteworthy development in recent years is the emergence of "immune checkpoint inhibitor" and "ipilimumab" between 2020 and 2022, according to the results of burst analysis of reference co-citations (**Table 2, Figure 9**). Immune checkpoint inhibitors (ICIs) have revolutionized cancer treatment, advancing drug therapy for malignant tumors. Unfortunately, KTRs are often excluded from clinical trials, resulting in limited data on the safety and efficacy of ICIs in this patient group. As a result, the use of ICIs off-label among KTRs with cancer is increasing based on case reports and small case series [41]. Although treatment with immune checkpoint inhibitors (ICIs) may be associated with improved outcomes for both patients and malignancies, there have been significant reports of transplant rejection, particularly in the case of combination therapy involving pembrolizumab and ipilimumab compared to monotherapy [42]. Therefore, it is crucial to conduct future large-scale clinical studies to develop personalized therapies for MKT. This should include assessing the safety and efficacy of ICIs in this specific patient population, to improve outcomes for patients, malignancies, and allografts. Such studies should follow a balanced approach, considering adverse events and prolonged survival [43].

This study presents a comprehensive overview of global research trends in the field of MKT. Initially, we employed bibliometric methods to thoroughly evaluate the current status and evolution of MKT-related studies. However, it is important to note that several limitations in our analysis may have introduced bias. For instance, we excluded literature from additional databases such as PubMed, Embase, Scopus, etc. Additionally, documents published after 2023 and non-English literature were also excluded from our analysis.

Conclusions

Considering the increased risk of malignancy among renal transplant recipients, current research focuses on developing early screening and prevention strategies for MKT and effective management of KTRs. Additionally, emerging research topics in the MKT field include ICIs and COVID-19 vaccines. In the future, it is necessary to strengthen international cooperation, gain deeper insights into MKT's risk factors and pathogenesis, and develop personalized screening, prevention, and treatment strategies, especially adopting modern treatment concepts for MKT to improve the long-term survival rate of KTRs.

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Declaration of Figures' Authenticity

All figures submitted have been created by the authors, who confirm that the images are original with no duplication and have not been previously published in whole or in part.

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